

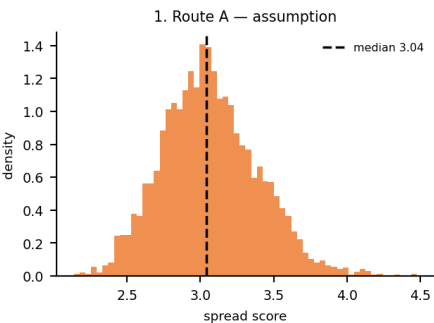
The Undocumented Pathogen — Two Routes

What can honestly be claimed about a disease with no data

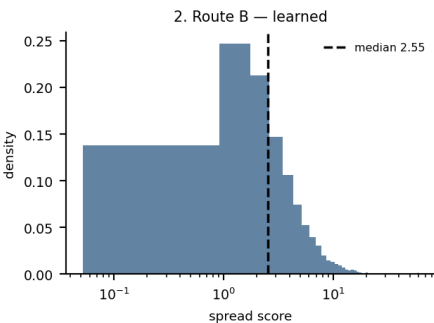
NO SYNTHETIC OBSERVATIONS ARE CREATED ANYWHERE IN THIS ARM. Neither route invents, simulates or imputes a single case. Route A propagates declared assumptions; Route B borrows from three measured outbreaks through the shared prior.

The two routes

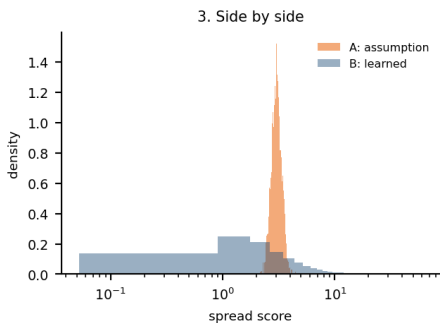
	Route A — assumption-driven	Route B — pooling-driven
What it does	push chosen parameters through the model	draw a new arm from the prior the measured outbreaks fitted
Observations used	none	717 across three outbreaks
Learns from data	no — posterior equals prior	yes — inherits the fitted hierarchy
Median spread score	3.04	2.55
89% interval	[2.57, 3.60]	[0.60, 10.81]
Honest claim	IF a pathogen behaved this way, THEN...	GIVEN three measured human outbreaks, a fourth would plausibly fall here
Not a claim	'the spread score is this number'	anything specific to this hypothetical pathogen



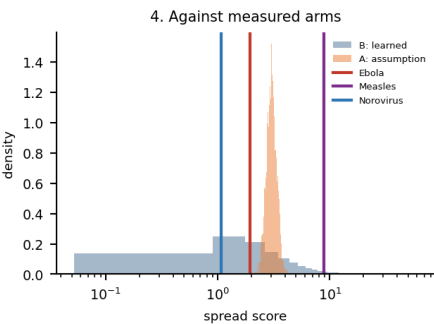
Route A: the assumption restated. Narrow because you chose it to be narrow — the width reflects your confidence, not any evidence.



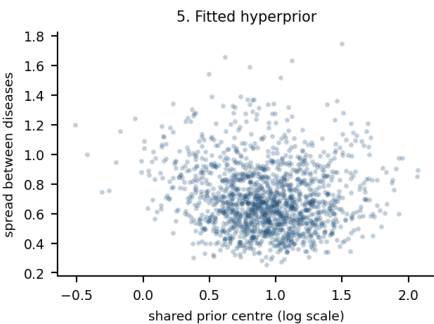
Route B: a new arm predicted from the fitted hierarchy. Wide because three measured outbreaks genuinely disagree about how transmissible pathogens are.



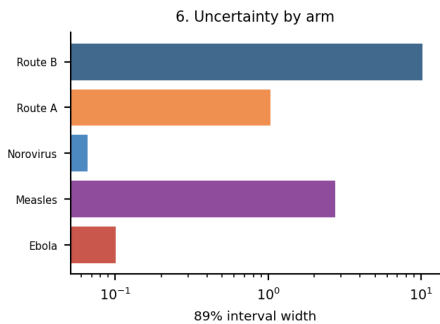
Both routes on one axis. Route A is a spike of your own choosing; Route B is a broad range shaped by data. They answer different questions.



Both routes against the three measured arms. Route B brackets them, as it must — it is the distribution they were drawn from.

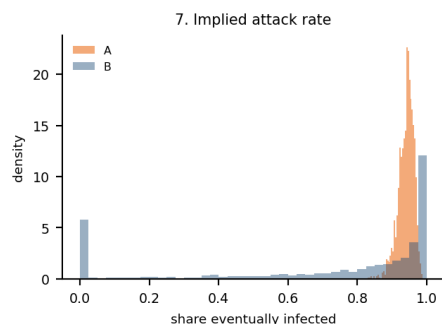


The shared prior the measured arms produced. Route B draws from exactly this, which is why it is data-informed.

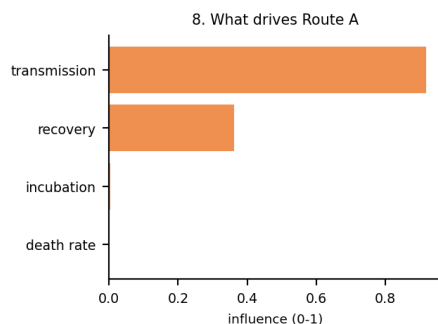


Interval width. Route B is far wider — honest uncertainty about an unmeasured pathogen, rather than borrowed confidence.

Implications and limits



Attack rate implied by each route. Route A insists nearly everyone is infected; Route B admits a wide range including mild outcomes.



Which assumption drives Route A. Transmission rate accounts for almost all of it; if data ever arrived, measure that first.

9. Information sources

Route A
assumptions -> ODE -> output
observations used: 0

Route B
Ebola + Measles + Norovirus
-> shared prior -> new arm
observations used: 717

Where each route's information comes from. Route A adds no data; Route B inherits three measured outbreaks.

What Route B is not

Route B is not a statement about zombies. The model holds no zombie information and cannot acquire any. It describes the population of human pathogens that Ebola, Measles and Norovirus were drawn from, and predicts where a fourth member of that population would fall. Presenting it as a pathogen-specific estimate would repeat Route A's error one level up.

THE SHORT ANSWER — the framework does not create information about an undocumented disease. It makes assumptions explicit, shows what they imply, identifies which one matters, and offers a second route where measured arms inform the unmeasured one. Route B is the defensible answer to 'can this handle unknown diseases'. Route A is a restatement and must be labelled as such.

Numeric comparison

Arm	Median	89% interval	Width	Relative spread	Note
Ebola	1.934	[1.88, 1.99]	0.10	2%	measured arm: informed by its own likelihood
Measles	8.857	[7.68, 10.43]	2.75	10%	measured arm: informed by its own likelihood
Norovirus	1.076	[1.05, 1.11]	0.07	2%	measured arm: informed by its own likelihood
Zombie — Route A	3.043	[2.57, 3.60]	1.03	11%	assumption restated; posterior equals prior; learns nothing from data
Zombie — Route B	2.551	[0.60, 10.81]	10.21	145%	new-arm prediction from the fitted hierarchy; informed by the measured arms